

[45, 3, 6, 7, 8, 1, 200, 100]

compare(a,b)

45, 5
3 6

(a,b) => a-b

6 7
45 7



AVL tree



JSON → String representation of Javascript object, is called as JSON

JSON.stringify(ob)	Convert javascript object into JSON format
JSON.parse(jsonob)	Convert JSON data into javascript object

Types of function

Self calling function	If a function calls itself, then it is called as self calling function. This function can be called only once
Closure function	A function, which is a nested function or child function is called as closure function This function has access to variables of parent function
Callback function	If we pass a function as a parameter to another function, and this function gets called after the main function finishes execution, then it is called as callback function But if lot of nesting of callback function is there, the code will be difficult to write and lot of changes need to be done

Geolocation API

Navigator.geolocation.getCurrentPosition(showposition,showerror)	getCurrentPosition is asynchronous function, it will call callback function showposition, once it finishes successfully, But if it fails, then it calls callback function showerror
Navigator.geolocation.watchPosition(showposition,showerror)	It is similar to getCurrentPosition, but watch position will give you live location
Navigator.geolocation.clearWatch	It will stop execution of watchPosition

Storage API

In Javascript there are 2 objects, localStorage and sessionStorage. These objects are browser specific

sessionStorage stores properties till you close the browser window

localStorage stores the properties, till we delete it explicitly

localStorage.getItem(name)	To retrieve value from locaStorage
localStorage.setItem(name,val)	To assign value to a property and save it
localStorage.removeItem(name)	To delete one of the property and its value from locaStorage/sessionStorage
localStorage.clear()	To delete all properties and its values

Promise object

To avoid callback hell, we use promises:-- when we use callback functions and if there is lot of nesting of callback function , then writing code will be tedious, this is called as callback hell

It is a object, which is in one of the three states

1. Pending, 2. Fulfill(resolved) 3. Rejected

Pending ->function is in progress

Resolve-> completed successfully

Rejected-> completed unsuccessfully

How to create promise object

```
Var p1=new Promise((resolve,reject)=>{
```

```
Var n=34;
```

```
If(n>10)
```

```
    resolve("valid number")
```

```
else
```

```
    reject( "invalid number")
```

```
})
```

```
P1.then((result)=>{
```

```
    Console.log("result found",result)
```

```
    Return 100;
```

```
})
```

```
.then((result)=>{
```

```
    Console.log(result))
```

```
.catch((err)=>{
```

```
    Console.log("error occurred",err)
  })
```

AJAX

AJAX(Asynchronous Javascript and XML)

1. When we want to replace portion of the page then we use AJAX
 - a. In a form if we select country, then the dropdown of state should get populated
 - b. After login to the site, it shows welcome username

Every asynchronous function return data via promise object